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Desenvolvimento Mobile - Android

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Broadcast Receivers

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Broadcast Receiver

- É comum em um dispositivo mobile captar **mensagens externas** para executar ações;
- Em um cenário compartilhado diversos eventos ocorrem e é importante **filtrar** os eventos.

Broadcast Receiver

Intent
Battery Low

Intent
Wi-Fi availability

Intent
Incoming Call

Intent
Bluetooth device connected

Intent
Incoming SMS

Intent
Charger (dis)connected

Broadcast Receiver

```
Intent broadcastIntent = new Intent();
broadcastIntent.setAction(Commons.BROADCAST_MESSAGE);
broadcastIntent.putExtra(Commons.BROADCAST_MESSAGE_CONTENT, "Broadcast Data");
sendBroadcast(broadcastIntent);
```

Broadcast Receiver

```
IntentFilter intentFilter = new IntentFilter();
intentFilter.addAction(Commons.BROADCAST_MESSAGE);
registerReceiver(receiver, intentFilter);
```

Broadcast Receiver

```
BroadcastReceiver receiver = new BroadcastReceiver() {
    @Override
    public void onReceive(Context context, Intent intent) {

        if (intent.getAction().equals(Commons.BROADCAST_MESSAGE)){
            //((TextView) findViewById(R.id.content)).setText(intent.getStringExtra(Commons.BROADCAST_MESSAGE_CONTENT));
            Toast.makeText(context, intent.getStringExtra(Commons.BROADCAST_MESSAGE_CONTENT), Toast.LENGTH_SHORT).show();

            Intent stopIntent = new Intent(context, BroadcastService.class);
            stopService(stopIntent);

            unregisterReceiver(this);
        }
    }
};
```

Comunicação com API REST

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Protocolo HTTP

Protocolo HTTP

Endpoint	Método	Ação
/users	GET	Retorna a lista de usuários
/users	POST	Insere um novo usuário
/users/{id}	GET	Retorna o usuário com id = {id}
/users/{id}	PUT	Substitui os dados do usuário com id = {id}
/users/{id}	PATCH	Altera items dos dados do usuário com id = {id}
/users/{id}	DELETE	Remove o usuário com id = {id}

API REST (Voley)

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Voley

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OVERVIEW | GUIDES | REFERENCE | SAMPLES | DESIGN & QUALITY

User data & identity
User location
Touch & input
CameraX
Camera
Sensors
Connectivity
Overview
Performing network operations
Transmit network data using Volley
Overview
Send a simple request
Set up RequestQueue
Make a standard request
Implement a custom request
Perform network operations using Cronet
Build client-server applications with gRPC
Transferring data without draining the battery
Reduce network battery drain
Transfer data using Sync Adapters

Volley overview

Volley is an HTTP library that makes networking for Android apps easier and most importantly, faster. Volley is available on GitHub.

Volley offers the following benefits:

- Automatic scheduling of network requests.
- Multiple concurrent network connections.
- Transparent disk and memory response caching with standard HTTP cache coherence.
- Support for request prioritization.
- Cancellation request API. You can cancel a single request, or you can set blocks or scopes of requests to cancel.
- Ease of customization, for example, for retry and backoff.
- Strong ordering that makes it easy to correctly populate your UI with data fetched asynchronously from the network.
- Debugging and tracing tools.

Voley

```
dependencies {
    ...
    implementation 'com.android.volley:volley:1.1.1'
}
```

Voley

```
// Instantiate the RequestQueue.
RequestQueue queue = Volley.newRequestQueue(this);
```

Voley

```
// Request a string response from the provided URL.
StringRequest stringRequest = new StringRequest(Request.Method.GET, url,
            new Response.Listener<String>() {
    @Override
    public void onResponse(String response) {
        // Display the first 500 characters of the response string.
        textView.setText("Response is: "+ response.substring(0,500));
    }
}, new Response.ErrorListener() {
    @Override
    public void onErrorResponse(VolleyError error) {
        textView.setText("That didn't work!");
    }
});
```

Voley

```
// Add the request to the RequestQueue.
queue.add(stringRequest);
```